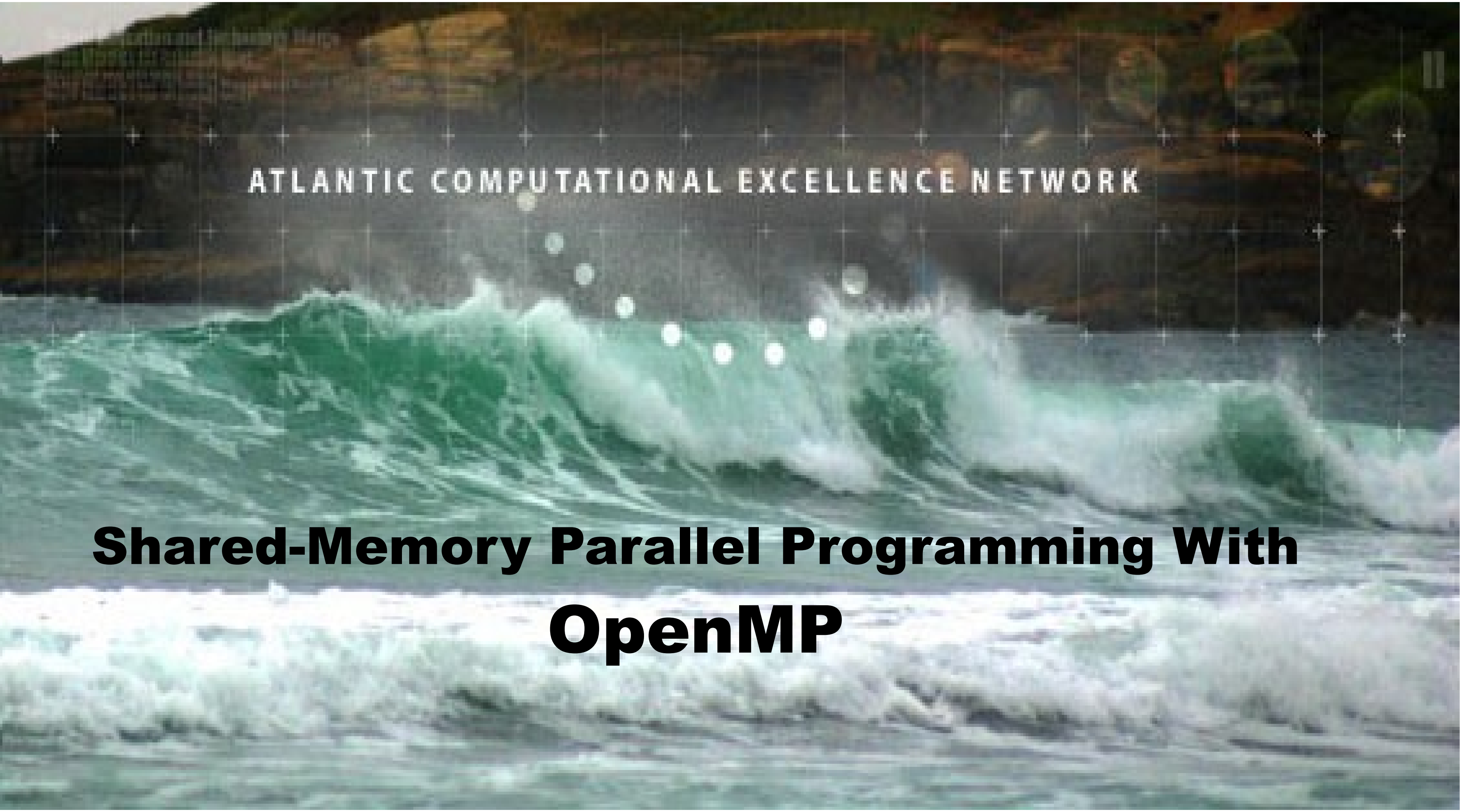

Shared-Memory Parallel Programming With OpenMP

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What is OpenMP?

- Open Multiprocessing API
  - First published 1997, v.3.0 published 2008
- A standard for shared-memory programming
  - with support from: AMD, Cray, Fujitsu, HP, IBM, Intel, NEC, Portland Group, Oracle, MS, TI; also ANL, LLNL, EPCC, NASA...
  - Fortran, C and C++ specifications in standard
- Shared-memory ≈ inside one multi-core machine
- Allows incremental parallelization

A Simple Loop

saxpy.f90

```
subroutine saxpy(z, a, x, y, n)
integer i, n
real z(n), a, x(n), y
!$omp parallel do
do i = 1, n
   z(i) = a * x(i) + y
end do
return
end
```

- Most work done with compiler directives:
  - `!$omp` *`directive`* in Fortran
  - `#pragma omp` *`directive`* in C, C++

Compilation

- OpenMP supported by these compilers at ACEnet:
  - Portland Group (PGI)
  - SunStudio
  - GNU version 4 and later
- Different flags for each compiler suite:

```
$ pgf90    -mp              saxpy.f90
$ f90      -xopenmp -O3 saxpy.f90
$ gfortran -fopenmp         saxpy.f90
```

Execution

- Environment variable OMP_NUM_THREADS controls number of threads

```
$ export OMP_NUM_THREADS=3
$ ./hello-c
Hello World from thread 0
Hello World from thread 2
Hello World from thread 1
There are 3 threads
$
```

bash syntax shown; in tcsh use “setenv OMP_NUM_THREADS 3”

Use with Grid Engine

- Best practice: Set OMP_NUM_THREADS from SGE variable NSLOTS

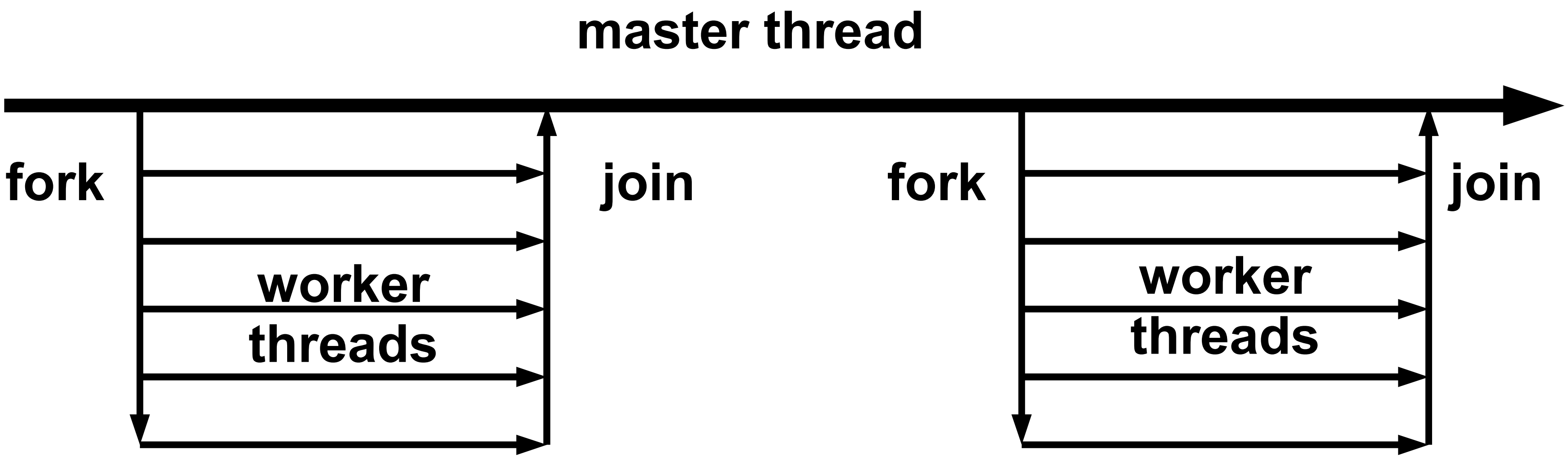
job.sh

```
#$ -S /bin/bash
#$ -l h_rt=0:1:0,test=true
#$ -pe openmp 4
export OMP_NUM_THREADS=$NSLOTS
./hello-c
```

Key Concepts

- Threads, fork, join
- Shared vs. private variables
- Reduction
- Data dependence
- Beyond loops: Parallel regions
- Scheduling

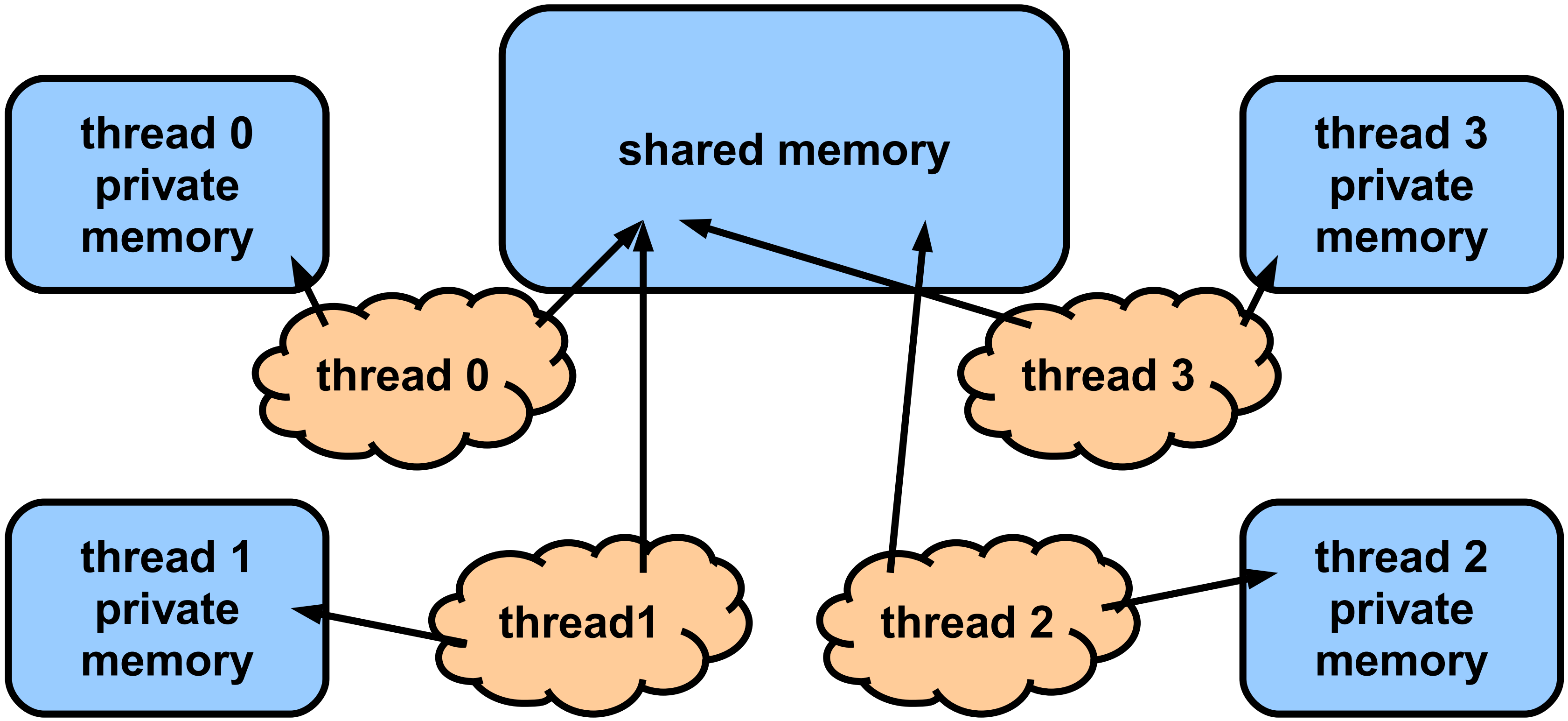
Threads



- Master thread “forks” (=creates) worker threads when code enters a parallel section
- Threads “join” (vanish) on leaving a parallel section

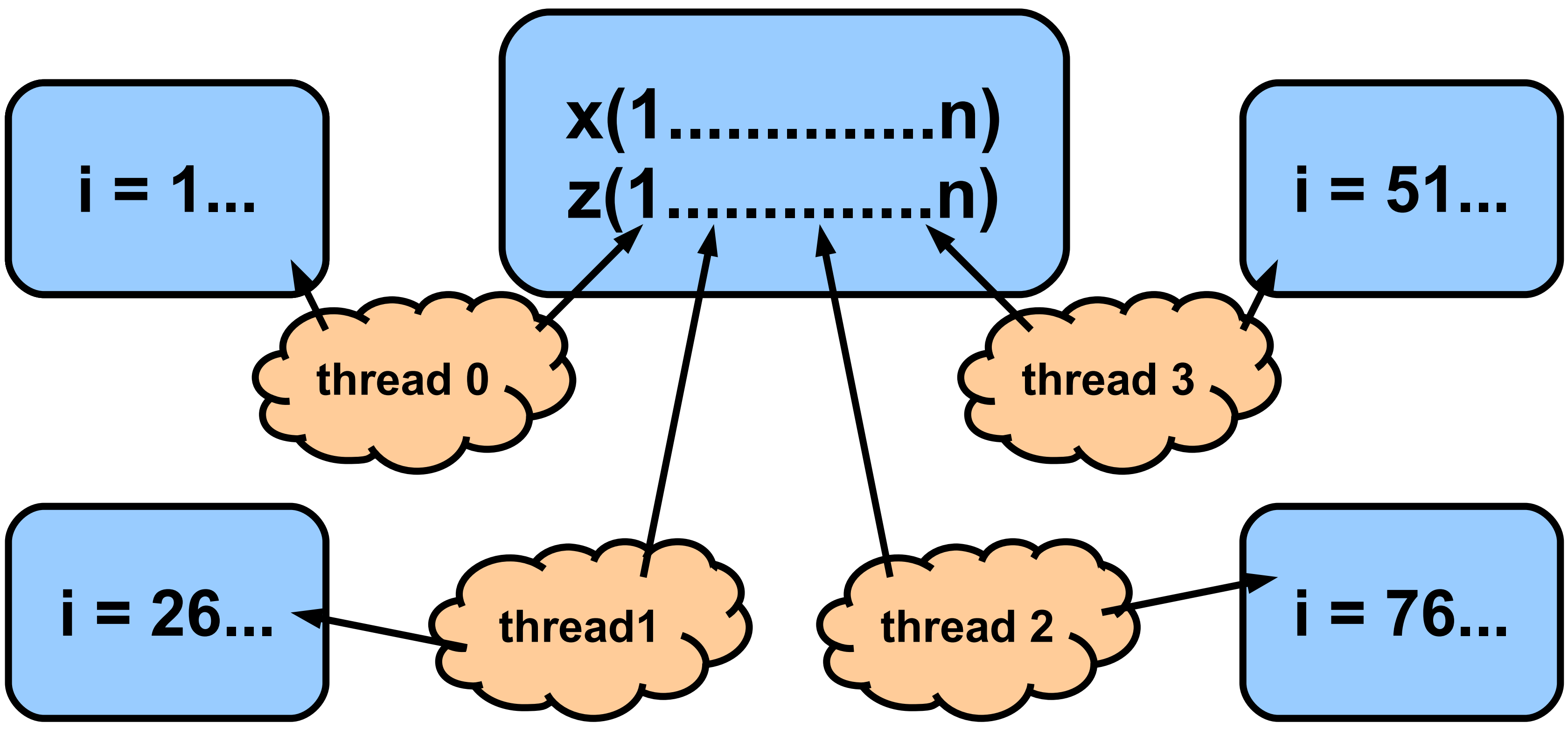
Shared vs. private memory

- Whole program has a shared memory area
- Each thread created with private memory



Shared vs. private memory

- Consider "saxpy" example:
  Array index "i" must be private to each thread

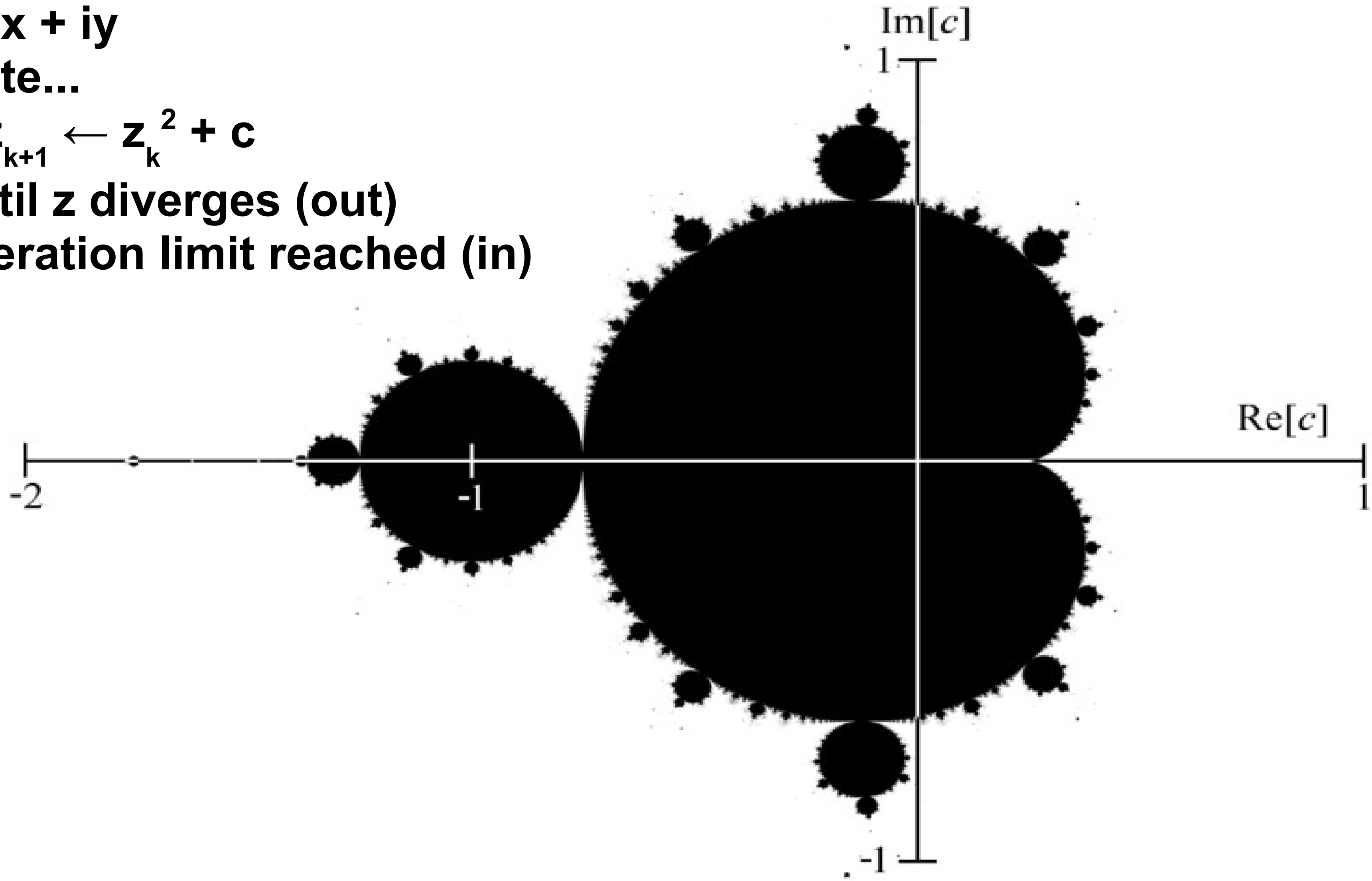


Shared vs. private variables

- There are default rules for scope
- Most variables shared by default
  - except Fortran loop indices
  - C “for” construct is more complicated
- Most private vars must have scope declared
  - Function (“stack”) locals are private
  - You *can* scope everything explicitly – maybe a good idea?
- See reference materials for details

Mandelbrot set example

For each point (x, y):
$c \leftarrow x + iy$
Iterate...
$z_{k+1} \leftarrow z_k^2 + c$
...until z diverges (out)
or iteration limit reached (in)



Nested loops

mandel.f90

```
!$omp parallel do private(j, x, y)
do i = 1, m
    do j = 1, n
        x = xmin + i*(xmax-xmin)/m
        y = ymin + j*(ymax-ymin)/n
        depth(j, i) = mandel_val(x, y, maxiter)
    end do
end do
```

- Usually best to parallelize outermost loop → 'i' private
- What about 'j' ?
- 'j' must be private too
- as must 'x' and 'y'
- 'mandel_val' should be a pure function

Reduction

reduce.f90

```
real function fmax(x, n)
integer i, n
real x(n)
fmax = -1.0d20
!$omp parallel do reduction(max:fmax)
do i = 1, n
   fmax = max(fmax, f(x(i)))
end do
return
end
```

- Shared variable used for global sum (product, max, min, logical *or*, bitwise *or*, logical *and, etc.* …)
- OpenMP controls access to reduction variable so threads don't suffer race condition

Data Dependence

If one statement writes a memory location,
and another statement either reads or writes the same location,
then there is a data dependence between the two statements.

- The order in which they are executed affects program correctness...
- ...and therefore affects parallelizability.

Data Dependence examples

```
do i = 2, n
    a(i) = a(i) + a(i-1)
end do
```

- Cannot parallelize this loop:

  a(i-1) must be *written* by iteration (i-1)
  before it is *read* in iteration (i).

```
do i = 2, n, 2
    a(i) = a(i) + a(i-1)
end do
```

- Can parallelize this loop:

  Even-numbered elements written;
  odd-numbered elements read – no dependence

Data Dependence examples

- What about this one?

```
do i = 2, n
    a(idx(i)) = a(idx(i)) + b(idx(i))
end do
```

- Depends on array **`idx`**
  - If it's a *permutation*, then no dependence
  - If it contains duplicate entries, then not okay
  - Compiler can't tell!
- Also: Beware common, module, global or static vars inside functions & subroutines!

Parallel Regions

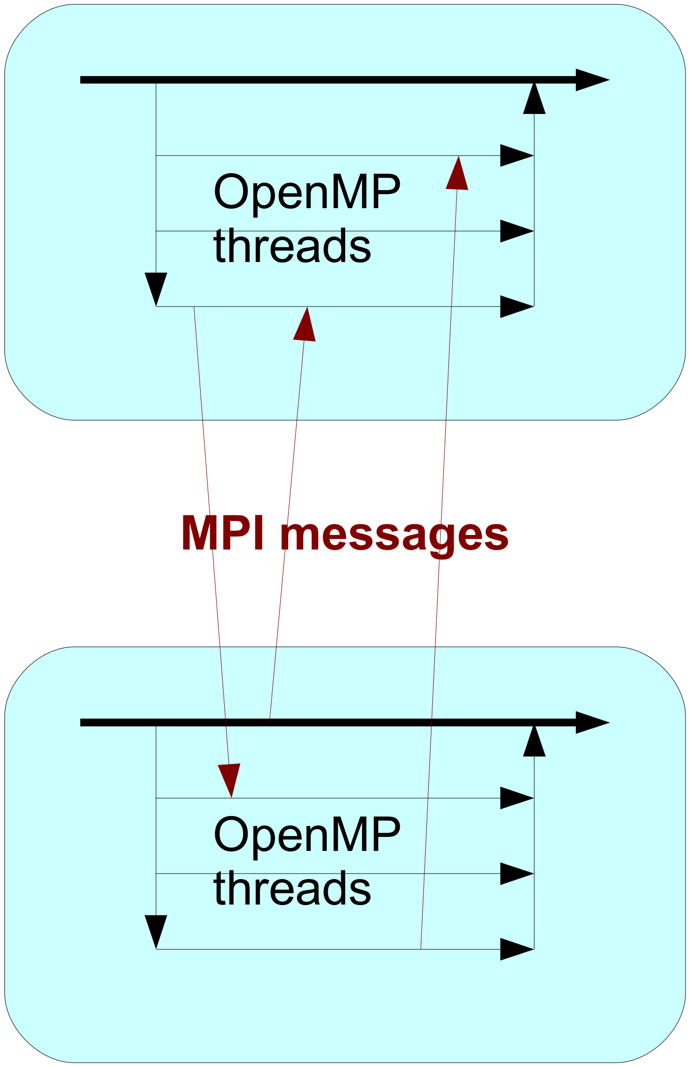
```
integer myid, nthreads
nthreads = omp_get_num_threads()

!$omp parallel private(myid)

myid = omp_get_thread_num()
call do_different_things( myid, nthreads )

!$omp end parallel
```

- Can parallelize general code, not just loops
- Probe OpenMP with functions

Not to mention...

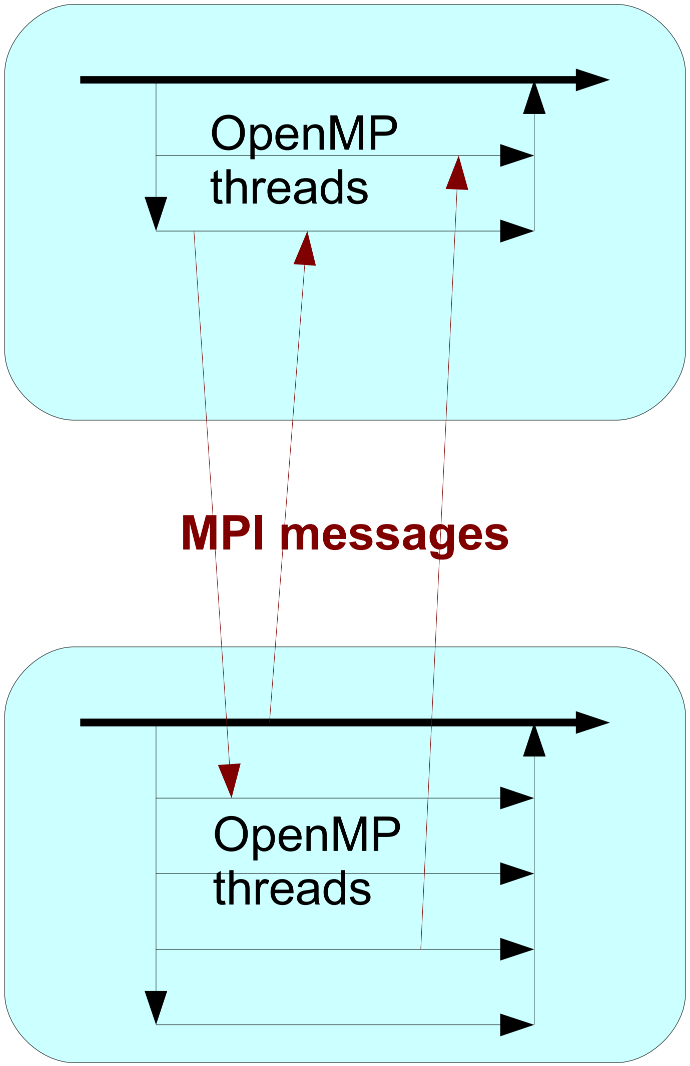
- Types of data dependence
    - and ways to fix some of them
- firstprivate, lastprivate, copyin
- do if, nowait
- Scheduling
    - static, dynamic, guided
- Synchronization
    - !$omp critical, barrier, atomic, ordered
- Work-sharing
    - !$omp sections, single, master

OpenMP + MPI



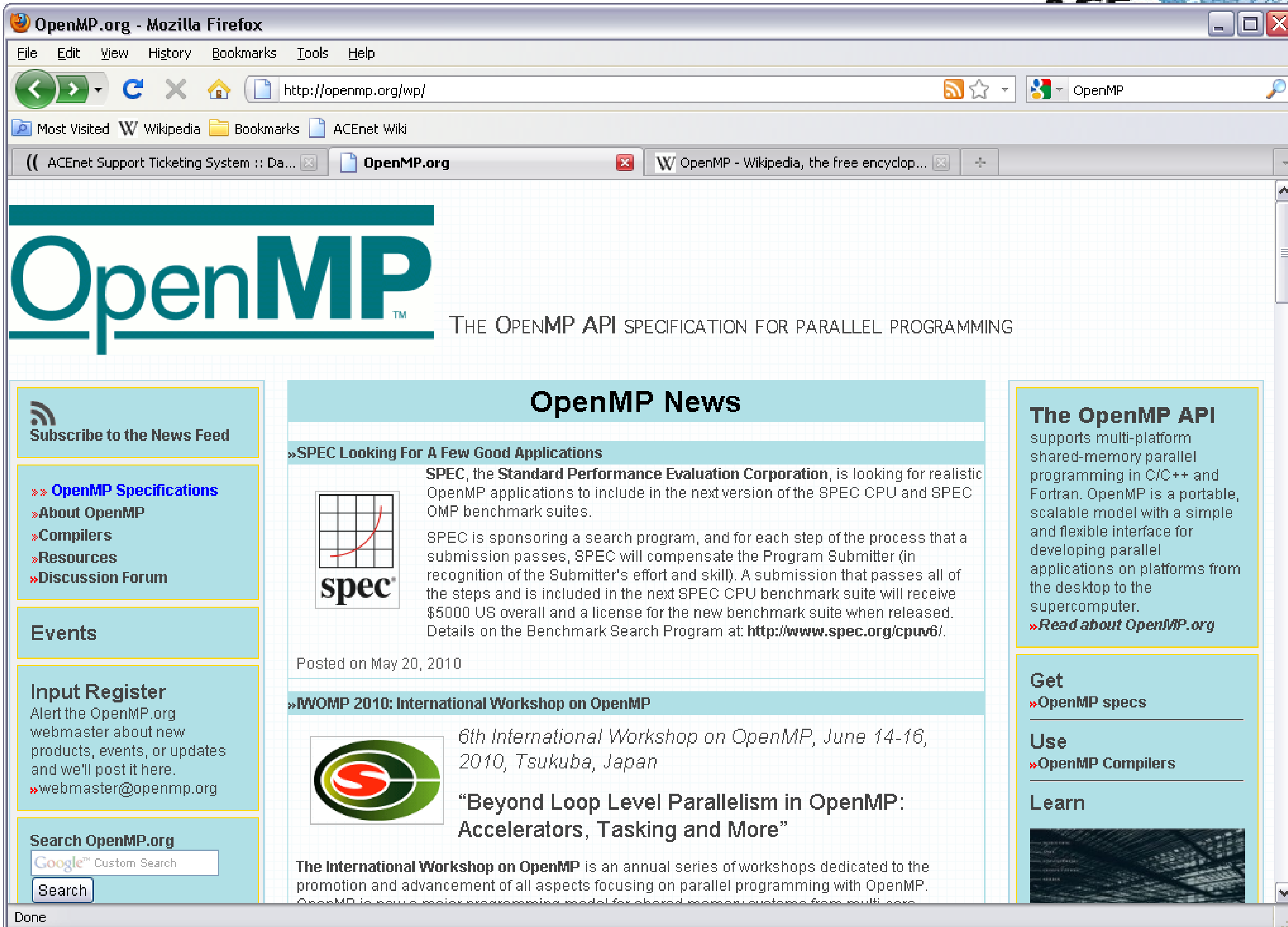
- Hybrid code easy to write
- *Can* give better performance in some cases;
  depends on the details
- See Quinn for examples
- But...

OpenMP + MPI



- ...interface with scheduler makes things tricky at ACEnet
- Grid Engine may give different number of slots on different hosts
- Code must handle this, *or*...
- … use unadvertised:
  `-pe twoper 8`
  `-pe fourper 8`

http://openmp.org



Reference materials

- http://www.openmp.org
- http://wiki.ace-net.ca for local documentation
- Examples drawn from Rohit Chandra *et al., Parallel Programming in OpenMP* (Academic Press, 2001)
- Michael J Quinn, *Parallel Programming in C with MPI and OpenMP* (McGraw-Hill, 2004)
- Example code on Brasdor, Courtenay, Fundy, Glooscap, Placentia in /home/rdickson/public/OpenMP_demo.tar